

Kyle Stellmann

Stellm98@rowan.edu | (973) 903-6047 | Portfolio: KyleStellmann.github.io/Portfolio

EDUCATION

Rowan University

Glassboro, NJ

Bachelor of Mechanical Engineering | 3.65/4.00

Expected May 2028

Honors & Distinctions

- President's Scholar of Excellence
- Dean's List

Fall 2024

Spring 2025 - Spring 2026

ENGINEERING PROJECTS

Underwater Remotely Operated Vehicle (UROV)

- Executed SolidWorks Flow Simulation (CFD) on hull geometry to evaluate drag coefficients, optimizing structural contouring to reduce total drag by 15% and increase top speed by 0.3 m/s.
- Developed close-form dynamic models balancing 10N motor thrust against hydrodynamic resistive force equations to calculate theoretical terminal velocity limits.

Precision Load Cell Scale & Embedded System

- Precision-machined an aluminum load flexure via lathe and mill to $\pm 0.010''$ tolerances, preventing plastic deformation under 0-5 kg maximum operational loads.
- Programmed an Arduino Nano and HX711 ADC in C++ to implement digital noise filtering and sensor calibration, integrating PCB power electronics inside a water-resistant CAD enclosure.

Stirling Engine Design and Prototyping

- Modeled complete assembly in SolidWorks, managing geometric constraints and calculating flywheel rotational inertia to ensure correct phase timing and continuous rotation.
- Tested and iterated interference fits from 3D-printed PLA-to-PVC to maintain pressure seal.

New Jersey Department of Military Affairs (NJDMA) Audit

- Evaluated 36 months of utility metrics against Heating/Cooling Degree Days (HDD/CDD) to build a weather-normalized facility baseline and pinpoint inefficiencies.
- Audited on-site HVAC systems, delivering a technical Energy Conservation Measure (ECM) report with cost-benefit analysis and calculated payback periods.

PROFESSIONAL ORGANIZATIONS

Martinson Honors College | Member

Spring 2025 - Present

American Society of Mechanical Engineers | Member

Fall 2026 – Present

CO-CURRICULAR ACTIVITIES

Assistive Technologies | Member

Fall 2024 – Present

- Developed low-actuation mechanical switch interfaces and structural safety restraints focused on human-factors controls and vehicle ergonomics.

WORK EXPERIENCE

Liscio's Bakery | Customer Service Associate

September 2025 – Present

- Delivered high-volume customer service, enforced quality standards, and optimized store workflow in a fast-paced retail environment.